

Mihaela-Elena Jipa

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EDUCATION

University POLITEHNICA of Bucharest, Faculty of Automatic Control and Computer Science 2024 – 2028

BSc Computer Science and Engineering

- Overall GPA (2nd year): **8.80**
- Relevant Coursework: DSA, Operating Systems, OOP, x86 Assembly, Communication Protocols

PROJECTS

PhishSense Jan 2026

Python-based phishing detection platform using Machine Learning

- Developed a **phishing detection platform** in **Python** capable of analyzing **URLs, emails, and QR codes** using **machine learning** and heuristic-based detection techniques.
- Trained and integrated two **machine learning models** for phishing URL and email detection, while building the full analysis pipeline with **Flask**, QR decoding (**pyzbar/Pillow**), persistent scan logging, and **Docker**-based deployment support.

Minimal ELF Loader Nov 2025

C application for loading and executing statically-linked ELF binaries on Linux

- Implemented a **custom ELF loader** in **C**, capable of loading and executing **statically-linked PIE and non-PIE binaries** on Linux, handling **virtual memory mapping, memory protection, relocation**, and setting up the **process stack (argc, argv, envp, auxiliary vectors)** for correct execution.

CopyServ Website Aug 2025

Responsive company website

- Built a **responsive website** for CopyServ using **HTML** and **CSS**, implementing a clean and modern design with intuitive navigation and layout.
- Deployed the site on **GitHub Pages** with a custom domain and configured **Cloudflare** to enable **HTTPS**, optimize performance, and enhance security.

TECHNICAL ACTIVITIES

ARM Summer School July 2026

- Participated in the ARM Summer School, July 2026, gaining hands-on experience with ARM-based embedded systems, the boot process (TF-A and U-Boot), Buildroot, embedded Linux, ARM TrustZone/TEE, and Linux device driver development on the NXP i.MX93 platform

MongoDB's Women in Tech Summit May 2026

- Participated in a selective, invite-only professional development summit focused on technical and career growth for women in tech.
- Attended expert-led sessions on technical interviewing, cloud technologies, and database management.
- Expanded technical knowledge of NoSQL, cloud computing, and emerging innovations in the data industry.
- Networked with engineers and leaders shaping the future of database technology.

Unikraft Open Source Contribution Dec 2025

- Contributed to the **Unikraft** open source project under mentorship by updating documentation for Python 3.14 support, integrating a Python library, and adding practical examples.

CONTESTS

ACS Keysight CTF — 4th Place May 2026

- Achieved 4th place in a cybersecurity Capture The Flag competition focused on security analysis, exploitation, cryptography, and reverse engineering.

HackITall 2025 – 3rd Place in SAP's Challenge Dec 2025

Developed a **custom heuristic optimization algorithm** for routing rotables on aircraft, minimizing total cost, and built a **dashboard** for visualizing airport stocks, penalties, and flight loads, while integrating **scheduling constraints** and **real-time data** with the organizers' evaluation platform.

TECHNICAL SKILLS

Programming & Scripting : Intermediate : **Python, C/C++, Java**

Beginner :HTML, CSS, Racket, Haskell, MATLAB, Assembly

Tools & Technologies **Git, Docker, Flask, Linux**

Languages **Romanian (native), English (C2)**